

As a first application of our new concepts, let us now extend the tree packing theorem (2.4.1) of Nash-Williams and Tutte to locally finite graphs. Its naive extension, with ordinary spanning trees, fails. Indeed, for every  $k \in \mathbb{N}$  one can construct a  $2k$ -edge-connected locally finite graph that is left disconnected by the deletion of the edges in any one finite circuit (Exercise 21). Such a graph will have at least  $k(\ell - 1)$  edges across any vertex partition into  $\ell$  sets, but it cannot have more than two edge-disjoint spanning trees: adding an edge of one of these to another creates a (finite) fundamental circuit there, whose deletion would disconnect any third spanning tree.

As soon as we replace ordinary spanning trees with topological ones, however, Theorem 2.4.1 does extend:

**Theorem 8.6.9.** *The following statements are equivalent for all  $k \in \mathbb{N}$  and connected locally finite multigraphs  $G = (V, E)$ :*

$G = (V, E)$

- (i)  $G$  has  $k$  edge-disjoint topological spanning trees.
- (ii) For every finite partition of  $V$ , into  $\ell$  sets say,  $G$  has at least  $k(\ell - 1)$  cross-edges.

We begin our proof of Theorem 8.6.9 with a compactness extension of the finite theorem. This yields a weaker, ‘finitary’, statement at the limit (cf. Lemma 8.6.7):

**Lemma 8.6.10.** *If for every finite partition of  $V$ , into  $\ell$  sets say,  $G$  has at least  $k(\ell - 1)$  cross-edges, then  $G$  has  $k$  edge-disjoint spanning submultigraphs whose closures in  $|G|$  are topologically connected.*

*Proof.* Pick an enumeration  $v_0, v_1, \dots$  of  $V$ . For every  $n \in \mathbb{N}$  let  $G_n$  be the finite multigraph obtained from  $G$  by contracting every component of  $G - \{v_0, \dots, v_n\}$  to a vertex, deleting any loops but no parallel edges that arise in the contraction. Then  $G[v_0, \dots, v_n]$  is an induced submultigraph of  $G_n$ . Let  $\mathcal{V}_n$  denote the set of all  $k$ -tuples  $(H_n^1, \dots, H_n^k)$  of edge-disjoint connected spanning submultigraphs of  $G_n$ .

(2.4.1)  
(8.1.2)

Since every partition  $P$  of  $V(G_n)$  induces a partition of  $V$ , since  $G$  has enough cross-edges for that partition, and since all these cross-edges are also cross-edges of  $P$ , Theorem 2.4.1 implies that  $\mathcal{V}_n \neq \emptyset$ . As every  $(H_n^1, \dots, H_n^k) \in \mathcal{V}_n$  induces an element  $(H_{n-1}^1, \dots, H_{n-1}^k)$  of  $\mathcal{V}_{n-1}$ , the infinity lemma (8.1.2), yields a sequence  $(H_n^1, \dots, H_n^k)_{n \in \mathbb{N}}$  of  $k$ -tuples, one from each  $\mathcal{V}_n$ , with a limit  $(H^1, \dots, H^k)$  defined by the nested unions

$$H^i := \bigcup_{n \in \mathbb{N}} H_n^i[v_0, \dots, v_n].$$

These  $H^i$  are edge-disjoint for distinct  $i$  (because the  $H_n^i$  are), but they need not be connected. To show that they have connected closures,

it suffices by Lemma 8.6.7 to show that each of them has an edge in every finite cut  $F$  of  $G$ . Given  $F$ , choose  $n$  large enough that all the edges of  $F$  lie in  $G[v_0, \dots, v_n]$ . Then  $F$  is also a cut of  $G_n$ . Now consider the  $k$ -tuple  $(H_n^1, \dots, H_n^k)$  which the infinity lemma picked from  $\mathcal{V}_n$ . Each of these  $H_n^i$  is a connected spanning submultigraph of  $G_n$ , so it contains an edge from  $F$ . But  $H_n^i$  agrees with  $H^i$  on  $\{v_0, \dots, v_n\}$ , so  $H^i$  too contains this edge from  $F$ .  $\square$

**Lemma 8.6.11.** *Every connected standard subspace of  $|G|$  that contains  $V$  also contains a topological spanning tree of  $G$ .*

*Proof.* Let  $X$  be a connected standard subspace of  $|G|$  containing  $V$ . Then  $G$  too must be connected, so it is countable. Let  $e_0, e_1, \dots$  be an enumeration of  $E(X)$ , and consider these edges in turn. Starting with  $X_0 := X$ , define  $X_{n+1} := X_n \setminus e_n$  if this keeps  $X_{n+1}$  connected; if not, put  $X_{n+1} := X_n$ . Finally, let  $T := \bigcap_{n \in \mathbb{N}} X_n$ .

Since  $T$  is closed and contains  $V$ , it is still a standard subspace. And  $T$  has an edge in every finite cut of  $G$ , because  $X$  does and its last edge in that cut will never be deleted. So  $T$  is connected, by Lemma 8.6.7. But  $T$  contains no circle: that would contain an edge, which should have got deleted since deleting an edge from a circle cannot destroy connectedness.  $\square$

**Proof of Theorem 8.6.9.** The implication (ii)  $\rightarrow$  (i) follows from our two lemmas. For (i)  $\rightarrow$  (ii), let  $G$  have edge-disjoint topological spanning trees  $T_1, \dots, T_k$ , and consider a partition  $P$  of  $V$  into  $\ell$  sets. If there are infinitely many cross-edges, there is nothing to show; so we assume there are only finitely many. For each  $i \in \{1, \dots, k\}$ , let  $T'_i$  be the multigraph of order  $\ell$  which the edges of  $T_i$  induce on  $P$ .

To establish that  $G$  has at least  $k(\ell - 1)$  cross-edges, we show that the multigraphs  $T'_i$  are connected. If not, then some  $T'_i$  has a vertex partition crossed by no edge of  $T_i$ . This partition induces a cut of  $G$  that contains no edge of  $T_i$ . By our assumption that  $G$  has only finitely many cross-edges, this cut is finite. By Lemma 8.6.7, this contradicts the connectedness of  $T_i$ .  $\square$

## (1.9) 8.7 The topological cycle space

As a more comprehensive application of our new theory, let us now look at how the cycle space theory of finite graphs extends to locally finite graphs  $G = (V, E)$  with infinite circuits and topological spanning trees.

Every two points of a topological spanning tree  $T$  are joined by a unique arc in  $T$ : existence follows from Lemma 8.6.5, while uniqueness is proved as for finite graphs. Adding a new edge  $e$  to  $T$  therefore creates